# Recurrence-free unconstrained handwritten text recognition using gated fully convolutional network

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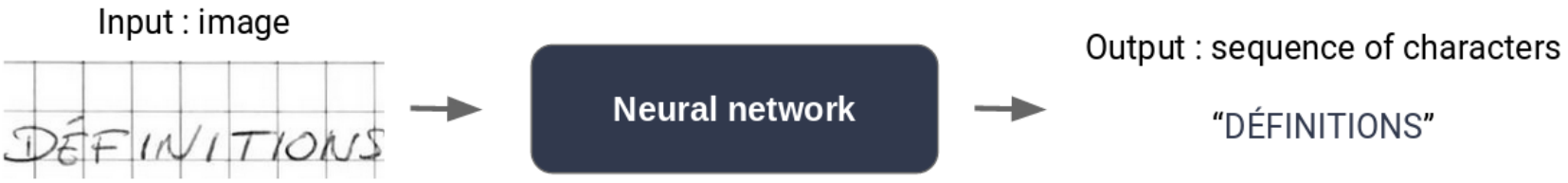
# Deep learning handwriting recognition system




## Constraints

- Images (input) of variable size
- Sequence of characters (output) of variable length

## Sequence alignment

Connectionist Temporal Classification (CTC) [Graves2006]

# State of the art

## Recurrent models (recurrent layers)

- Multi-Dimensional Long-Short Term Memory (MDLSTM) [Pham2014; Voigtlaender2016]
- Convolutional Neural Network + Bidirectional Long-Short Term Memory (CNN+BLSTM) [Puigcerver2017]

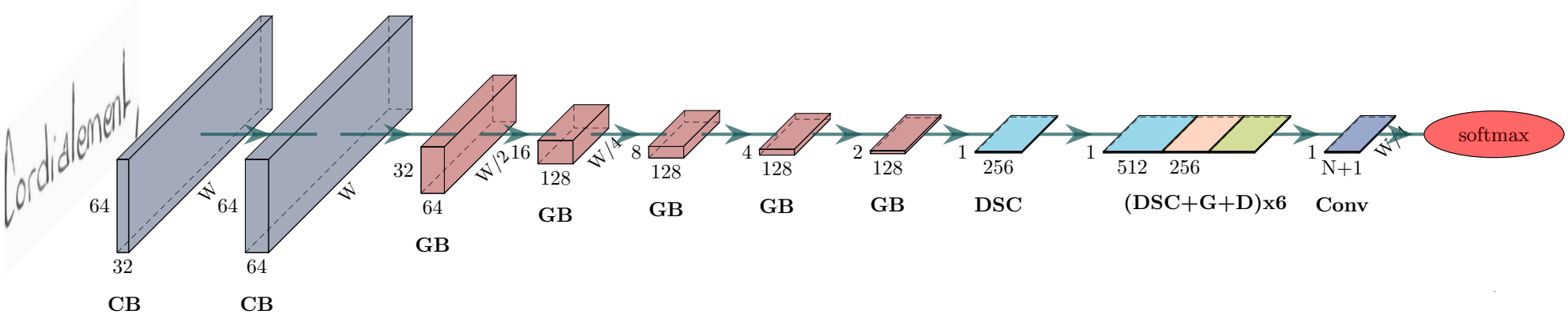
## Non-recurrent models

- Convolutional Neural Networks (CNN) [Ptucha2018]
- Gated Fully Convolutional Network (GFCN) [Yousef2018; Ingle2019]

## Attention models (recurrent process)

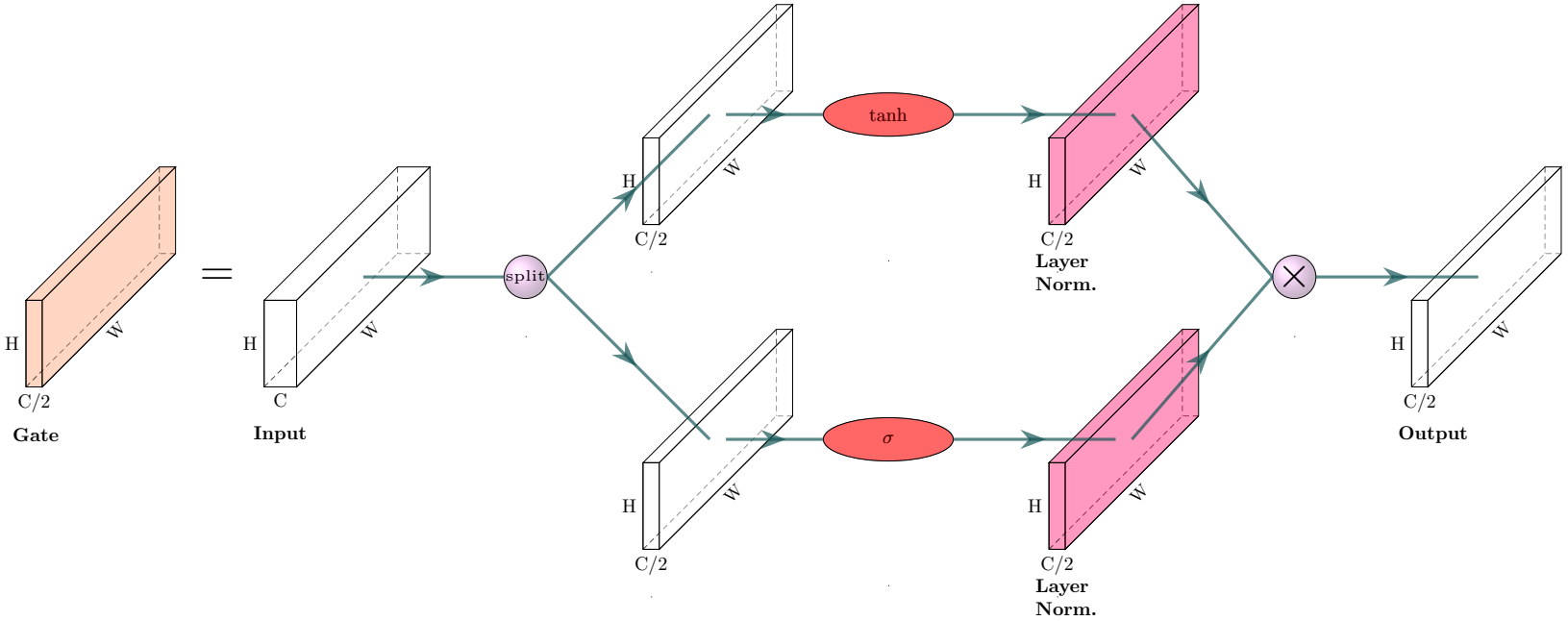
- Encoder-decoder architecture with soft attention [Chowdhury2018; Michael2019]

# Our GFCN model




- G (Gate)
- DSC (Depthwise Separable Convolution)
- CB (Convolution Block) = Conv + Conv + Instance Norm. + Dropout
- GB (Gate Block) = DSC + DSC + Instance Norm. (+ MaxPooling) + Gate + Dropout
- (H, W, C) = (Height, Width, Feature maps)
- N = charset size (+ 1 for the CTC blank)

# Gate


H
W
C/2
Gate
=
H
W
C
Input
split
H
W
C/2
tanh
H
W
C/2
Layer
Norm.
H
W
C/2
σ
H
W
C/2
Layer
Norm.
H
W
C/2
Output

# Details

## Architecture

- Deep: 22 convolutional layers
- Parameters: 1.4 M
- Receptive field: (196, 240)
- Input: Fixed-height image (64px) preserving the original width

## Hyperparameters

- Framework: Pytorch
- Optimizer: Adam(0.0001)
- loss: CTC

# Datasets

- grayscaled line text images (300dpi)

## Dataset characteristics

| Dataset | Training | Validation | Test | Alphabet | Language |
|---|---|---|---|---|---|
| RIMES | 9,947 | 1,333 | 778 | 100 | French |
| IAM | 6,482 | 976 | 2,915 | 79 | English |

## Example

[RIMES]


[IAM]

# Normalization techniques - visualisation

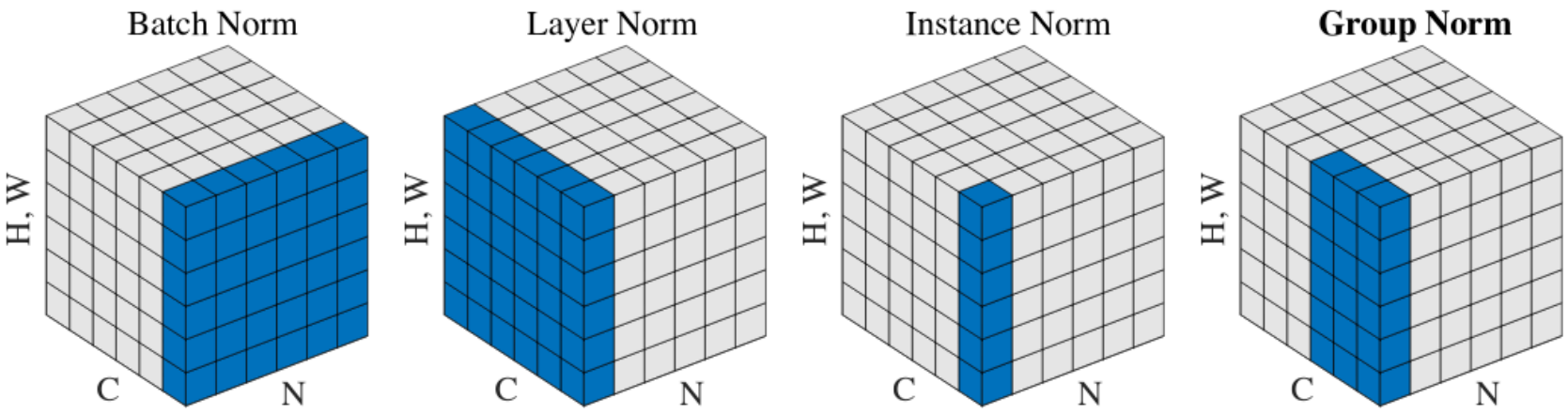



Image from [Wu2018]

- N: Mini-batch size
- H: Height
- W: Width
- C: Feature maps

# Normalization techniques - results

| Normalization | CER (%) 50 epochs | CER (%) 100 epochs | CER (%) 150 epochs | CER (%) 200 epochs | Time (/epoch) |
|---|---|---|---|---|---|
| Instance | 6.87 | **5.03** | **4.47** | **4.28** | **8.5 min** |
| Layer | **6.75** | 5.04 | **4.47** | **4.28** | 15 min |
| Group (32) | 7.10 | 5.30 | 4.86 | 4.32 | 8.75 min |
| Batch | 9.6 | 5.7 | 5.4 | 4.8 | **8.5 min** |

Table: Effect of type of normalization for our GFCN with the RIMES dataset (for a mini-batch size of 2). CER is computed on the valid set.

# Impact of ending blocks

| Number of ending blocks (DSC+G+D) | CER (%) 100 epochs | CER (%) 200 epochs | Parameters | Receptive Field (h, w) |
|---|---|---|---|---|
| 6 (baseline) | 6.82 | **5.80** | 1,375,792 | (196, 240) |
| 5 | **6.69** | 5.97 | 1,241,904 | (196, 212) |
| 4 | 8.14 | 7.48 | 1,108,016 | (196, 184) |
| 3 | 6.93 | 6.23 | 974,128 | (196, 156) |
| 2 | 7.35 | 6.63 | 840,240 | (196, 128) |
| 1 | 8.30 | 7.83 | 706,352 | (196, 100) |

Table: Impact of the receptive field on the IAM dataset. CER is computed over the valid set.

# IAM

| Architecture | CER (%) validation | WER (%) validation | CER (%) test | WER (%) test | Parameters |
|---|---|---|---|---|---|
| 2D-LSTM [Moysset2019] | 5.41 | 20.15 | 8.88 | 29.15 | 0.8 M |
| 2D-LSTM-X2 [Moysset2019] | 5.40 | 20.40 | 8.86 | 29.31 | 3.3 M |
| CNN + 1D-LSTM [Puigcerver2017] | 5.1 | | 8.2 | | 9.6 M |
| CNN + 1D-LSTM [Moysset2019] | **4.62** | 17.31 | **7.73** | 25.22 | 9.6 M |
| Ours | 5.23 | 21.12 | 7.99 | 28.61 | 1.4 M |

Table: Comparative results on the IAM dataset without LM, lexicon nor data augmentation.

# RIMES

| Architecture | CER (%) validation | WER (%) validation | CER (%) test | WER (%) test | Parameters |
|---|---|---|---|---|---|
| 2D-LSTM [Moysset2019] | 3.32 | 13.24 | 4.94 | 16.03 | 0.8 M |
| 2D-LSTM-X2 [Moysset2019] | 3.14 | 12.48 | 4.80 | 16.42 | 3.3 M |
| CNN + 1D-LSTM [Moysset2019] | **2.9** | 11.68 | 4.39 | 14.05 | 9.6 M |
| CNN + 1D-LSTM [Puigcerver2017] | 3.0 | | **3.3** | | 9.6 M |
| Ours | 3.82 | 15.60 | 4.35 | 18.01 | 1.4 M |

Table: Comparative results on the RIMES dataset without LM, lexicon, nor data augmentation.

# Conclusion

## Our model

- A recurrent-less fully convolutional network
- Deep, with a large receptive field
- Competitive results on both RIMES and IAM datasets

## Future works

Improving the performances

- Implement a data augmentation strategy

Toward paragraph-level text recognition

- Seq2Seq model with attention [Bluche2016; Bluche2017]

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[Yousef2018] M. Yousef et al. *Accurate, Data-Efficient, Unconstrained Text Recognition with Convolutional Neural Networks*. 2018.

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[Michael2019] Johannes Michael et al. *Evaluating Sequence-to-Sequence Models for Handwritten Text Recognition*. 2019. eprint: 1903.07377.

[Moysset2019] B. Moysset and R. Messina. "Are 2D-LSTM really dead for offline text recognition?" In: *IJDAR* 22 (June 2019), pp. 1–16. DOI: 10.1007/s10032-019-00325-0.

# Standard Convolution

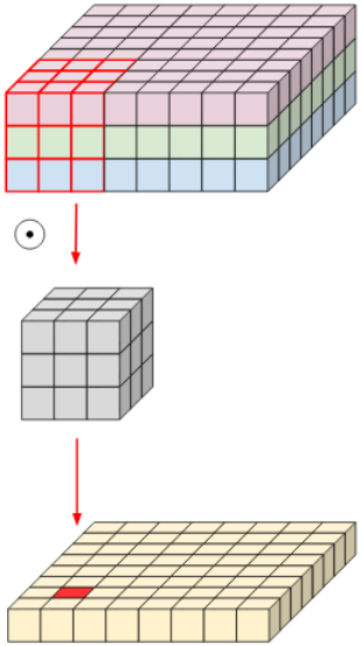
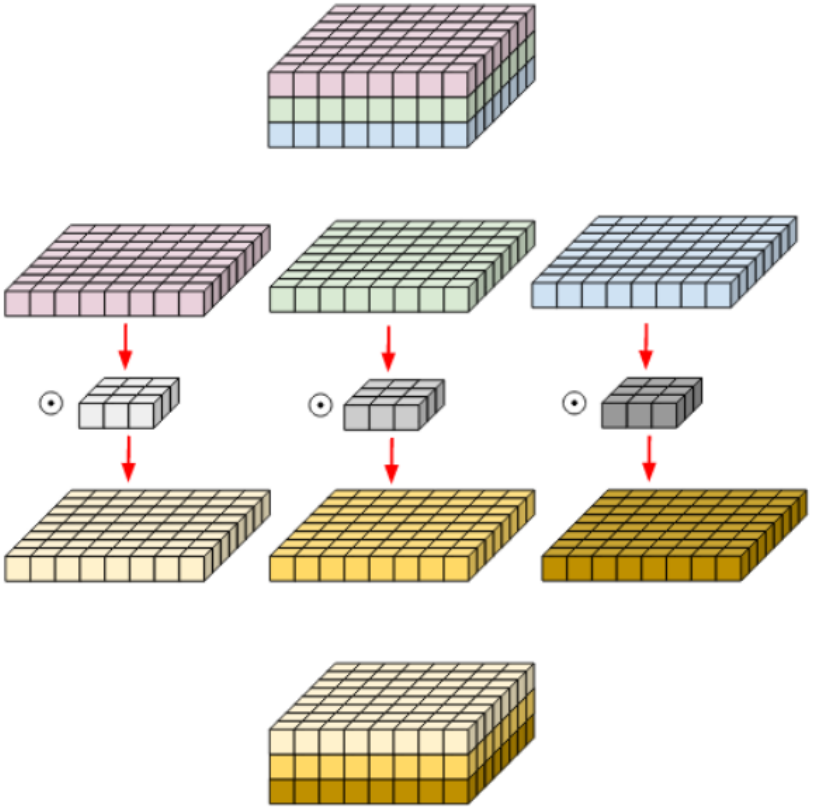
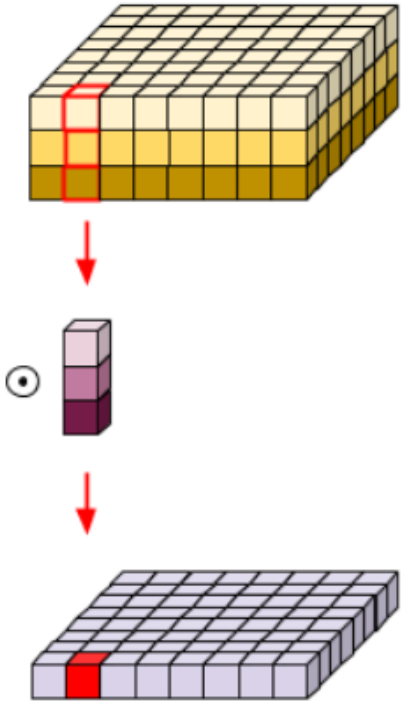


Image from
https://eli.thegreenplace.net/2018/depthwise-separable-convolutions-for-machine-learning/

# Depthwise Separable Convolution



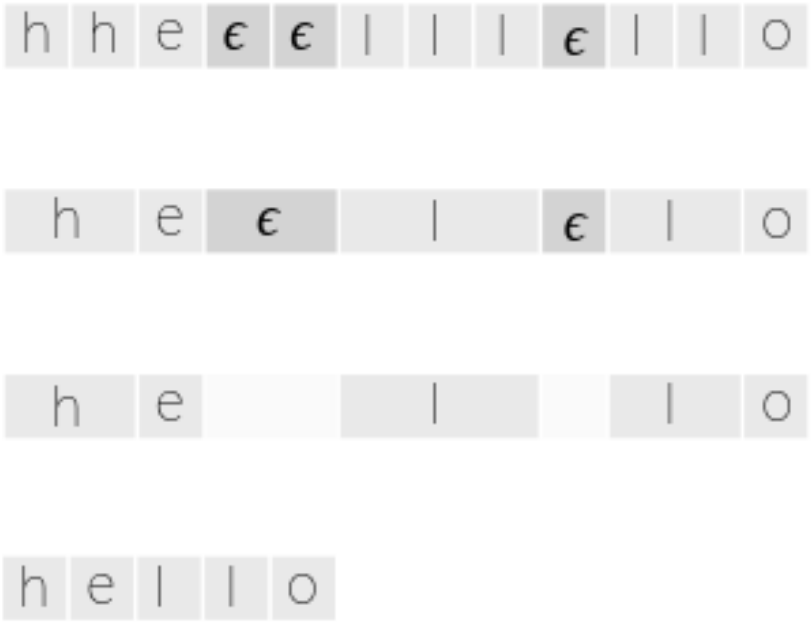
(a) Depthwise convolution



(b) Pointwise convolution

Image from
https://eli.thegreenplace.net/2018/depthwise-separable-convolutions-for-machine-learning/

# CTC




First, merge repeat characters.

Then, remove any $\epsilon$ tokens.

The remaining characters are the output.

Image from `https://distill.pub/2017/ctc/`

# Evolution since last year

| | GCNN (2019) | GFCN (2020) |
|---|---|---|
| RIMES Valid CER | 9.92 % | 3.82 % |
| RIMES Test CER | 10.03 % | 4.35 % |
| Convolutional layers | 21 | 22 |
| Parameters | 6.9 M | 1.4 M |
| Normalization | ✗ | Instance |
| Framework | Keras | Pytorch |
| Input image height | 32px | 64px |
| Keep ratio | ✓ | ✗ |
| Data Augmentation | ✓ | ✗ |
| Depthwise Separable Convolution | ✓ | ✓ |
| Residual connection | ✓ | ✗ |
| Shared-weight layers | ✓ | ✗ |
| Fully-connected layers | ✓ | ✗ |
| Sliding window process | ✓ | ✗ |